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Estimation of temperature using Numerical methods
Given:

$$\frac{dT}{dt} = -0.1(T-25)$$

Initial condition: $T_0 = 90^\circ\text{C}$

step size: $h = \frac{1}{3}$ hour

Number of steps 3

a) Euler's Method

Formula:

$$T_{n+1} = T_n + hF(T_n)$$

$$F(T) = -0.1(T-25)$$

$$\text{Step 1: } T_1 = 90 + \frac{1}{3} [-0.1(90-25)] = 90 - 2.167 = 87.833$$

$$\text{Step 2: } T_2 = 87.833 + \frac{1}{3} [-0.1(87.833-25)] = 87.833 - 2.094 = 85.739$$

$$\text{Step 3: } T_3 = 85.739 + \frac{1}{3} [-0.1(85.739-25)] = 85.739 - 2.025 = 83.714$$

Final Answer (~~Euler's~~)

$$T \approx 83.71^\circ\text{C}$$

(b) Modified Euler method (Heun's method)

$$T_{n+1} = T_n + \frac{h}{2} [F(T_n) + F(T_{\text{predict}})]$$

$$\text{predictor: } T_p = 90 + \frac{1}{3} [-0.1(65)] = 87.333$$

$$\text{corrector: } T_1 = 90 + \frac{1}{6} [-6.5 - 6.283] = 90 - 2.132 = 87.869$$

$$T_2 \approx 85.807$$

$$T_3 \approx 83.817$$

$$\text{Final } \boxed{T \approx 83.82^\circ\text{C}}$$

(c) Second order Runge-kutta method (midpoint method)

~~from~~

$$k_1 = hf(T_n)$$

$$k_2 = hf\left(T_n + \frac{k_1}{2}\right)$$

$$T_{n+1} = T_n + k_2$$

Step 1:

$$k_1 = \frac{1}{3} [-0.1(65)] = -2.167$$

$$k_2 = \frac{1}{3} [-0.1(63.917)] = -2.131$$

$$T_1 = 90 - 2.131 = 87.869$$

Step 2: $T_2 \approx 85.807$

Step 3: $T_3 \approx 83.817$

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$$\boxed{T \approx 83.82^\circ\text{C}}$$

$$\frac{8ms}{2314126}$$